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The effect of Chinese import competition on Mexican local labor markets

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ABSTRACT

This paper contributes to the literature of the effect of globalization on labor markets in developing economies by analyzing the Mexican case. I exploit variation across Mexican regions in import exposure stemming from initial differences in industry specialization in order to estimate the effect Chinese competition had on local Mexican labor markets. Also, by taking advantage of the Mexican exports' high dependence on the U.S. market, I estimate the effect that China-caused trade diversion had on Mexican labor markets. I find that the increase in competition decreased the employment share in manufacturing for the average Mexican local labor market. This effect was found to be larger for regions with high exposure to Chinese competition in the U.S. market, showing that there was a significant, negative indirect effect from China's trade growth. I also find that workers' mobility across local labor markets increased due to this negative shock. Wages remained largely unaffected.

1. Introduction

The effect of globalization on labor markets is a topic that has been studied for almost two decades now. Starting on mid-1990s, a great body of literature trying to explain what caused the marked changes in the U.S. wage structure during the 1980s and 1990s was amassed. The main findings were

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that, although significant, globalization in the form of trade and offshoring played a rather small role in the wage differentials in U.S. labor markets (Feenstra & Hanson, 1999; Feenstra & Hanson, 2003), with the evidence suggesting that other shocks, among which skill biased technological change stood out, were the main factors in the evolution of the U.S. wage structure in that decade (Katz & Autor, 1999). A somewhat stronger role for globalization in the rising wage inequality was found in later research involving developing countries, which in the 1980s began their opening process consisting mostly in the aggressive lowering of barriers to trade and capital flows. This openness was associated with increases in the relative demand for skilled labor in economies like Chile (Pavcnik, 2003) and Mexico (Feenstra & Hanson, 1997), with the resulting increase in wage inequality. However, as pointed out in Autor, Dorn, and Hanson (2013), a factor limiting trade's impact on labor markets is that, historically, imports from low-wage countries, a likely source of disruption to high-wage labor markets (Krugman, 2008), had been relatively small. Nonetheless, emerging economies with staggering rates of economic and trade growth, among which China stands out, have completely changed the picture. China's trade volume has grown dramatically over the last two decades. Between 1990 and 2010, China's share of the United States' total imports went from 3.1% to 18.4%. At the same time, China went from representing less than 1% of Mexico's total imports to accounting for 14% of them. Under these new circumstances, trade, particularly with low-income countries, seemed to have a clear potential to disrupt labor markets both for developed and developing countries.

Several factors account for China's trade growth: the transition to a market-oriented economy, which has involved over 150 million Chinese workers migrating from rural areas to cities; access to long banned foreign technologies, capital goods, and intermediate inputs; multinational firms allowed to operate in the country; and China's accession to the World Trade Organization. These developments made of China the primary supplier of labor-intensive type of goods to the North American market, replacing Mexico as the second biggest exporter to the United States. Several studies, some of which I mention below, have exploited these events to reassess the effect of trade on labor markets. However, recent events make us believe that China's export frenzy has reached its climax and it is now perhaps the best moment to assess its effects. Also, if, as forecasted by some studies, China's advantage is reverting, it is in the interest of every country to have an idea of what they can expect in the next few years. This is especially relevant to countries like Mexico, where the manufacturing industry is specialized in labor-intensive goods, industry in which China would be expected to experience the largest drawback. This study attempts to exploit the plausibly exogenous growth in China's exports into Mexico in order to estimate the causal impact of Chinese competition on key Mexican labor market outcomes.

My study is related to several areas of research. First, a few studies have assessed the threat that Chinese exports represent to Latin American producers. A couple of outstanding examples are Hanson and Robertson (2007, 2008) and Soloaga, Oarreaga and Lederman (2007). Other studies (Pavcnik and Goldberg, 2005 and Eslava et al., 2009 for Colombia and Brazil, respectively) evaluate the effect of Chinese import competition on wages and employment in Latin America. In a sectoral study, Iranzo and Ma (2006) find some evidence of Chinese exports diverting US imports from Mexico toward China. Some studies more closely related to mine have found evidence suggesting growth in China's trade affected negatively the performance of Mexico's manufacturing industry. For example, Shigeoka, Verhoogen, and Wai-Poi (2006) use plant-level data on Mexican maquiladora and non-maquiladora manufacturing sector to investigate the heterogeneous impact of the Chinese export expansion at the plant level within industries. They find that Chinese imports into Mexico of goods also produced by Mexican plants have a negative impact on those plants' sales. However, they leave out plenty of interesting outcomes such as wages within and outside of the manufacturing industry, employment in non-manufacturing industries, labor force mobility, among others. Also, by focusing on industry rather than on geographic subdivisions (e.g., local labor markets), they do not take advantage of variation by region in the exposure to the rise of China's trade. A very recent paper that fills some of these gaps is Autor, Dorn, and Hanson (2013). They explore the effect of import competition on U.S. local labor markets that were differentially exposed to the growth in China's trade between 1997 through 2007 due to differences in their initial patterns of industry specialization. They find that increased exposure of local labor markets to Chinese imports leads to higher unemployment, lower labor force participation, and reduced wages. They also find that exposure to Chinese trade did not have a significant effect on labor mobility. Iacovone at al. (2013) follow an approach similar to Autor et al. (2013) in order to estimate the effect of Chinese competition on Mexican firms. They find that while smaller plants' sales and marginal products are compressed and more likely to cease, those of larger firms are more resilient to the shock.

Similarly to Autor et al. (2013), in this study, I estimate the economic impact of Chinese import competition on local Mexican labor markets by exploiting the variation in industry specialization. Differences in the initial industry mix in a local labor market (municipality) imply different levels of exposure to import competition. For example, consider two different municipalities: A and B. In the year 2000, municipality A's most important industry was apparel manufacturing, an industry in which China has a strong competitive advantage. In the same year, municipality B's most important industry was hand gun manufacturing, an industry China is not very well known for. By the end of the decade, considering Mexican imports from China grew by a multiple of 10 from 2000 to 2010, municipality A experienced a substantial increase in exposure to Chinese import competition while municipality B did not. I would then expect to see a larger effect of import competition on municipality A's labor market outcomes.

One of the contributions of this paper is that, to the best of my knowledge, this is the first study on the effects of exogenous trade shocks on local labor markets for developing countries that does not involve the country under study voluntarily opening-up to trade and foreign investment (see, for example, Kovak, 2011). Also, taking local labor markets as the unit of analysis allows me to measure a broad set of economic impacts beyond wages and employment in manufacturing: employment and wages in the non-manufacturing sector, unemployment, labor-force participation, and labor mobility. Additionally, the data I had access to allows me to relax some assumptions made in previous studies (namely, Autor et al., 2013) and, in principle, improve upon their estimates. Finally, since Mexico has the peculiarity of exporting nearly 90% of its manufactures to the U.S., we can test with relatively weak assumptions whether Chinese exports, by diverting US imports from Mexico, have had an indirect effect on the Mexican economy.

The rest of the paper proceeds as follows. Section 2 describes the theoretical motivation for a region's exposure index to China's imports, which I use as the main explanatory variable in a semi-structural approach. Section 3 gives a detailed description of the data used for the study. Section 4 presents the empirical methodology followed. Results are presented in Section 5, and Section 6 concludes.

2. Theoretical motivation

I want to know how exogenous growth in Mexican imports from China (i.e., growth caused by China's increase in productivity or accession to the WTO) affects the demand for goods produced by Mexican local labor markets. In order to do this, and to motivate the empirical approach to determining the level of exposure to Chinese import competition, I borrow from the main result in the study by Autor et al. (2013). Under this framework, region *i*, the local labor market of interest, is treated as a small open economy within a monopolistic competition model. The task is then to derive how shocks in China affect region *i*'s employment and wages. Then, this estimate is used in a reduced form model to estimate the effect of the trade shock on the desired outcomes (which in this case are Mexican local labor market employment, wages, and workers' mobility). For the U.S. case, the result for traded goods in region *i* is:

$$\hat{L}_{Ti} = -\propto \sum_{j} \frac{L_{ij}}{L_{Ti}} \frac{X_{ijU}}{X_{ij}} \frac{M_{UjC}}{E_{Uj}} \hat{A}_{Cj} \tag{1}$$

in which employment in the traded sector in region *i* depends on the exogenous growth of U.S. imports from China (as captured by $M_{UjC}\hat{A}_{Cj}$), scaled by total expenditure in the U.S. on industry *j* (E_{Uj}), and weighted both by the share of industry *j* in region *i*'s total employment in the traded sector (L_{ij}/L_{Ti}) and by the share of output by region *i*, industry *j*, that is shipped to the domestic market (shipped within the U.S., as opposed to abroad), captured by X_{ijU}/X_{ij}. The main assumptions made to arrive to these results are: (i) there is a trade imbalance between the U.S. and China and (ii) China affects the U.S. regions only through greater import competition in the U.S. market.

This first set of assumptions is relatively weak, since U.S. imports from China vastly exceed U.S. exports to China, which suggests both that trade is imbalanced and that the export channel through which China might affect U.S. regions is small. I keep these assumptions in place for this study since, in these instances, the Mexican case is analogous to that of the U.S. However, the authors then proceed to make another set of considerably stronger assumptions. First, they make use of the fact that, in the monopolistic competition model with a single factor of production, L_{ij}/X_{ij} equals a constant. To the extent that L_{ij}/X_{ij} varies by region and/or industry, the instrument might be biased. They also assume that the share of region i in total U.S. purchases in industry j (X_{ijU}/E_{Uj}) can be approximated by the share of region i in U.S. employment in industry j (L_{ij}/L_{Uj}), an assumption motivated by their lack of data on regional production and sales. They recognize that variation in capital or skill by industry would create noise in the instrument. Thus, putting together all the assumptions, the authors end up with the following result for traded goods in region i:

$$\widehat{L_{Ti}} \approx -\tilde{\alpha} \sum_j \frac{L_{ij}}{L_{Uj}} \frac{M_{UjC}\hat{A}_{Cj}}{L_{Ti}} \tag{2}$$

in which employment in the traded sector in region i depends on the exogenous growth of U.S. imports from China (as captured by $M_{UjC}\hat{A}_{Cj}$), scaled by region i's labor force (L_{Ti}), and weighted by the share of region i in U.S. employment in industry j (L_{ij}/L_{Uj}). The change in the wage and the change in employment in the non-traded sector are defined analogously. In the case of this study, access to production and sales data at the municipality level allows me to avoid using the second set of assumptions, decreasing, in principle, the likelihood of bias and thus ensuring a better estimate of the effect of Chinese competition on region i. The theoretical result I use for the Mexican case would then be:

$$\hat{L}_{Ti} = -\propto \sum_j \frac{L_{ij}}{L_{Ti}} \frac{X_{ijm}}{X_{ij}} \frac{M_{mjC}}{E_{mj}} \hat{A}_{Cj} \tag{3}$$

where the index m stands for Mexico and the rest of the variables are defined analogously to the U.S. case, that is: $M_{UjC}\hat{A}_{Cj}$ captures the exogenous growth of Mexican imports from China, which are then scaled by total expenditure in Mexico on industry j (E_{mj}); this expression is then weighted by two additional terms: X_{ijm}/X_{ij}, the share of output by region i, industry j, that is sold within Mexico; and L_{ij}/L_{Ti}, the share of industry j in region i's total employment in the traded sector. The intuition behind Eq. (3) is straightforward: an exogenous arrival of Chinese output from industry j can only affect a labor market outcome in Mexican region i if (i) region i sells output from the same industry j in the domestic market (i.e., ships within Mexico), and (ii) employment in industry j represents a positive share of total employment in region i.

Again, the form in Eq. (3) does not include the effect of exports to China nor it includes the indirect effect on Mexico of Chinese exports into the U.S. The latter assumption will be relaxed to measure the magnitude of the effect that trade diversion had on Mexican labor markets, while the former will be kept in place throughout the analysis since, by 2010, only about 1% of Mexican exports had China as their destination. This is a semi-structural approach in the sense that I then proceed to use the resulting equation to, under some assumptions, estimate an index of exposure to Chinese imports and then use it as the main explanatory variable in a reduced-form approach.

3. Data

Fig. 1 shows Mexican imports from China between 1990 and 2010 (Source: UN Comtrade). In 1990 Mexico only imported a few hundred thousand dollars worth of goods from China (all values are expressed in 2010 prices). By the year 2000, this number had increased to nearly four billion dollars. However, it was not until the decade between 2000 and 2010 that we see a large and sustained growth rate in this variable, reaching nearly 50 billion dollars by 2010. For this reason, my analysis of the effect of Chinese exports on Mexican labor markets focuses on this decade.

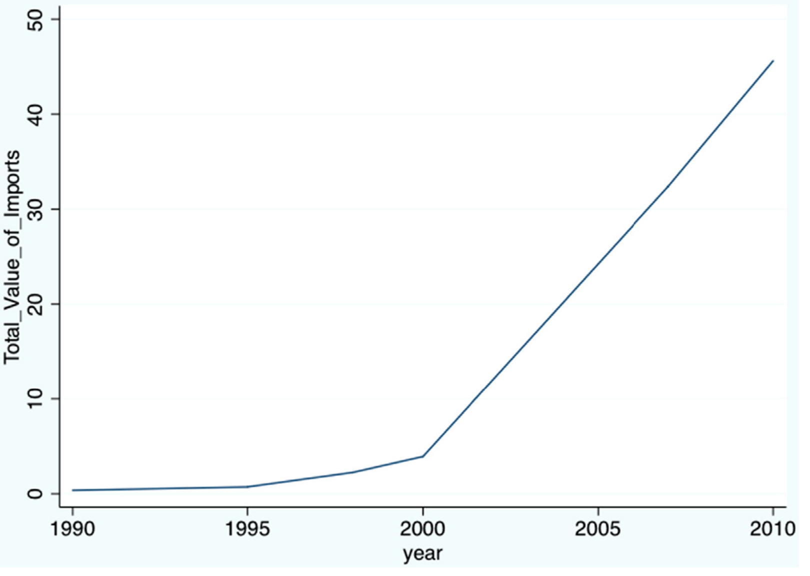

Fig. 1. Mexican imports from China 1990–2010 (in billions, 2010 US dollars).

In this study I use three main data sources: the UN Comtrade Database, the Mexican Economic Census, and the Mexican Population Census. I use data from the UN Comtrade Database on Mexico, U.S., and a set of both developed and developing countries' imports from China at the five-digit SITC product level. This level of disaggregation allows me to have the values of imports by importer for 180 manufacturing industries. Since I am interested in measuring the effect of the decadal change in local labor market exposure to Chinese products on several outcomes, I use data from the years 2000 and 2010. The measure for the exposure of local labor markets to imports from China combines trade data with information on local production, sales, and employment in detailed industries. Information on industry structure by local labor market is obtained from the Mexican Economic Census, a survey run every 5 years, which covers almost all economic activity in Mexican territory.[1] This survey provides me with data on production, sales, and employment by municipality and by detailed industry for the year 1998. To define my concept for local labor markets I then proceed to aggregate the data for those municipalities that, according to the 2005 INEGI Metropolitan Zones delimitation, have a high degree of socioeconomic integration.[2] The remaining municipalities are also included in the sample, although some of them are dropped in some specifications in order to test for heterogeneous effects. In total, I count 1403 local labor markets, which include both metropolitan zones and single municipalities. The main dependent variables studied in the empirical analysis are local labor market employment, wage, and population by education, age, and gender. All nominal values are adjusted to 2010 prices. These variables are constructed from the Mexican Population Censuses for the years 2000 and 2010, which were obtained from the Minnesota Population Center's Integrated Public Use Microdata Series (IPUMS), in its international version (2014).

For concreteness and clarity, I now make explicit the data that is used for each of the variables in Eq. (3). $\hat{L}_{Ti}$ is the change in employment in manufacturing in region i; L_{ij} is total manufacturing employment in region i in industry j; X_{ijm} represents sales in Mexico from region i in industry j; X_{ij} *is*

[1] The Economic Census excludes agriculture, stockbreeding, forestry, hunting, passenger transportation in collective automobiles of fixed route, taxis and limousines services, political associations, political organizations, international organisms, and organisms outside of Mexican territory.

[2] There are 56 Metropolitan Zones in the Mexican territory, covering 345 municipalities in 29 states, and accounting for 75% of the Gross Domestic Product. These zones include 56% of the national population and 78.6% of the urban population.

total sales from region i in industry j; M_{mcj} represents Mexican imports from China in industry j; and E_{mj} is total production in Mexico in industry j.

4. Empirical strategy

The empirical proxy to Eq. (3) is:

$$\Delta Exp_{mit} = \sum_j \frac{L_{ijt}}{L_{it}} \frac{X_{ijmt}}{X_{ijt}} \frac{\Delta M_{mjct}}{E_{mjt}} \tag{4}$$

where t denotes that the variable is measured at the beginning of the period, ΔExp_{mit} is the observed change in the level of exposure of Mexican municipality i between the start and the end of the period, and ΔM_{mjct} is the observed change in Mexican imports from China in industry j, also between the start and the end of the period. Expression (4) yields a measure of Chinese import growth per \$1000 U.S. dollars produced by region i. We can identify two sources of variation in this measure. First, differences by region in the manufacturing industry mix generate variation in the exposure level across regions. Second, the national share of manufacturing employment the region represents will also be reflected in differences in the estimate. However, in the main estimating equation I control for initial manufacturing employment, leaving the identification to come solely from variation in industry mix.

One major concern when estimating the level of exposure of Mexican local labor markets is that Mexican imports from China may be affected by Mexican demand shocks rather than just China's growing productivity and falling trade costs. In order to isolate the exogenous supply side growth of Chinese exports to Mexico, I follow a strategy analogous to the one in Autor et al. (2013). I instrument for Mexican imports from China using Chinese exports to other middle-income countries.[3] Additionally, anticipated growth in trade from China might affect the behavior of Mexican manufacturers, causing a bias that would attenuate the effect of import exposure on labor market outcomes. To mitigate this bias I use lagged levels of sales, output, and labor. Adding these two modifications to the instrument gives us

$$\Delta Exp_{oit} = \sum_j \frac{L_{ij-2}}{L_{i-2}} \frac{X_{ijm-2}}{X_{ij-2}} \frac{\Delta M_{ojct}}{E_{mj-2}} \tag{5}$$

where the main differences between this expression and the expression in (4) are: (i) ΔExp_{mit} and ΔM_{mjct} have been replaced by ΔExp_{oit} and ΔM_{ojct} respectively, denoting the use of imports from other middle-income countries rather than Mexico; and (ii) this expression uses levels of sales, output, and employment lagged by 2 years. To the extent that shocks to local Mexican labor markets stemming from internal shocks to product demand or technology are correlated across the middle-income countries used for the construction of the instrument, the instrumental variable could be biased. However, this correlation is likely to attenuate the negative effect of Chinese import exposure since positive demand shocks would typically increase both employment and Chinese imports. It seems plausible that correlated technology shocks would also attenuate the adverse effect of import competition on employment. A positive technology shock would typically increase employment and consumption (Christiano & Vigfusson, 2004), increasing also imports from China.

The regression model to be used throughout the analysis is:

$$\Delta y_{it} = \beta_1 \Delta Exp_{oit} + X'_{it}\beta_2 + e_{it} \tag{6}$$

where Δy_{it} is a 10-year change in employment, wages, or population; X'_{it} contains start of the period manufacturing employment share and demographics; and standard errors are clustered by state.

[3] The countries included in the sample are Argentina, Brazil, Chile, Colombia, Costa Rica, Greece, Panama, and Portugal, and were selected for their income proximity to Mexico based on income statistics presented in the World Bank report (2001).

Table 1
Imports exposure level summary.

Descriptive statistics for imports exposure per USD$1000 of output across MAs (in 2010 USD) Percentiles			
I. 2000		II.2010	
95th percentile	$13.40	95th percentile	$119.0
75th percentile	$12.50	75th percentile	$85.0
50th percentile	$5.00	50th percentile	$38.3
25th percentile	$2.24	25th percentile	$16.2
5th percentile	$0.30	5th percentile	$3.0

$N = 1387$. Observations weighted by start of the period (2000) working age population.

Table 2
Summary statistics of Mexican local labor markets.

	Levels		Decadal change
	2000	2010	2000–2010
Country-level imports from China to Mexico/(production) (in kUS$)	0.012	0.118	0.106
Country-level imports from China to Mexico/(workers) (in kUS$)	0.083	0.792	0.709
Average Import Exposure Index (ΔExp)	0.003	0.021	0.018
	(0.010)	(0.057)	(0.047)
% Working age population employed in manufacturing	13.30	11.23	−2.08
	(10.37)	(8.67)	(5.33)
% Working age population employed in non-manufacturing	30.75	36.32	5.57
	(13.52)	(14.26)	(6.49)
% Working age population unemployed	0.54	2.57	2.03
	(0.69)	(1.86)	(1.93)
% Working age population not in the labor force	48.06	46.21	−1.86
	(8.78)	(7.01)	(7.35)
% Working age population with college	4.037	7.79	3.42
	(3.82)	(5.68)	(2.70)
Log weekly wage in manufacturing	5.90	6.58	0.68
	(0.60)	(0.45)	(0.48)
Log weekly wage in non-manufacturing	5.98	6.63	0.64
	(0.50)	(0.42)	(0.34)

Unweighted averages, $N = 1402$ for both 2000 and 2010.

5. Results

Table 1 shows the variable Exp_{mit} for representative percentiles both at the beginning and at the end of the period. On average, the amount of incoming dollars from Chinese imports per $1000 dollars produced in the local labor market increased by a multiple of 8. For example, in the first column we can see that a region in the 75th percentile received only $12.5 dollars for every thousand dollars produced in 2000, but in 2010 that amount increased to $85 dollars, as shown in the second column. In Table 2 we can find summary statistics for the main characteristics of Mexican local labor markets. At the country level, Mexico received only $12 dollars in Chinese imports for every $1000 dollars produced in manufacturing in 2000; by 2010 this number had increased to $118 dollars. In a similar exercise, I find that while in the year 2000 Mexico imported only $83 dollars per worker, by the end of the decade imports from China per worker amounted to $792 dollars. Some preliminary results show that during the decade, on average, manufacturing employment decreased by 2 percentage points; employment

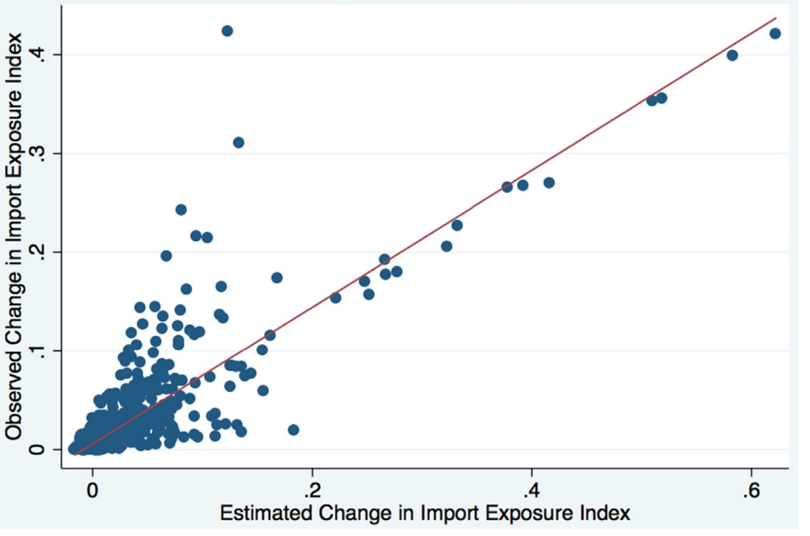

Fig. 2. First stage regression.

in the non-manufacturing industry[4] increased by 5.57 percentage points; and the weekly log wage slightly increased for both manufacturing and non-manufacturing industries.

5.1. Manufacturing employment

Fig. 2 sketches the first-stage regression of the two-stage least squares, which shows the strength of the relationship between the observed change in Chinese import competition faced by Mexican local labor markets and the estimated one using Chinese exports to other middle-income countries. There exists a very significant and strong relationship between the observed change in import exposure and the estimated one (F-statistic = 127). This should make us feel confident about the strength of the instrumental variable.

In Table 3 we can find the two-stage least square estimates for the change in manufacturing employment in local labor markets due to the increase in import exposure. Columns 1–4 show the instrumental variable estimate with different control variables. In column 1, I control only for beginning of the period proportion of manufacturing in total employment. The coefficient −17.1 is significant at the 1% level and it implies that an increase of one standard deviation in import exposure would result in a drop of 0.56 percentage points in manufacturing employment.[5] As I include the beginning of the period demographic control variables we can see a drop in the coefficient, while still remaining economically large and strongly significant. The preferred specification (column 4) includes as controls the beginning of the period proportion of manufacturing employment, proportion of workers with a college degree, proportion of women in the working population, and the average age of workers. The coefficient −13.7 implies that an increase of one standard deviation in import exposure would decrease manufacturing employment by 0.45 percentage points. As comparison, column 5 shows the ordinary least squares estimate. The OLS coefficient is 12.2% smaller than the IV one, hinting at a small attenuation effect, which suggests that, although present, endogeneity and pre-adjustment biases are small. Fig. 3 graphs the specification under column IV (4), which includes the full set of controls.

[4] The non-manufacturing industry includes all service industries and excludes Agriculture, fishing, and forestry, Mining, Electricity, gas, and water, and Construction.

[5] I calculate this by multiplying the coefficient by 0.033, which is the standard deviation of the level of exposure.

Table 3
Imports from China and change in manufacturing employment in municipalities.

Variable	2000–2010				
	IV (1)	IV (2)	IV (3)	IV (4)	OLS
ΔExp	−17.057 (4.802)**	−15.677 (4.425)**	−14.623 (4.562)**	−13.667 (4.540)**	−11.293 (3.778)**
L. % Emp Manuf	−0.216 (0.017)**	−0.218 (0.016)**	−0.217 (0.017)**	−0.224 (0.015)**	−0.225 (0.016)**
L. % College		−6.844 (2.285)**	−4.624 (3.736)	−4.54 (3.418)	−4.39 (3.403)
L. % Women			−3.822 (3.454)	−3.583 (3.260)	−4.343 (2.977)
L. Age				−0.378 (0.194)	−0.389 (0.198)
R^2	0.54	0.55	0.55	0.56	0.56
N	1387	1387	1387	1387	1387

All regressions include a constant. SE are clustered on state. Models are weighted by start of the period MA share of national working age population with college.
+ $p < 0.1$.
* $p < 0.05$.
** $p < 0.01$.

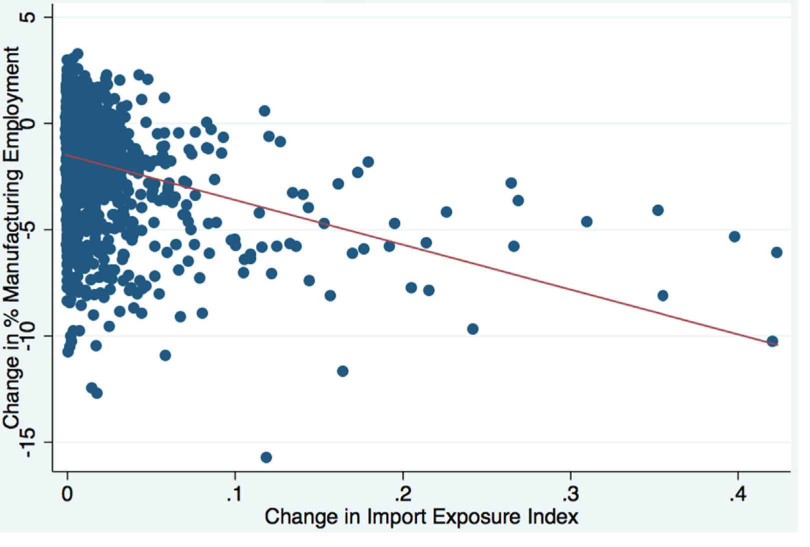

Fig. 3. Change in manufacturing employment, 2000–2010.

5.2. Other local labor market outcomes

One of the advantages of treating the local labor market as the unit of analysis is that it enables me to measure a broad set of outcomes outside of the manufacturing industry. If a country's labor markets are not geographically integrated, fully competitive, and in continuous equilibrium, then shocks to local manufacturing employment may also differentially affect employment, unemployment, and wages in the surrounding local labor market (Autor et al., 2013). It seems plausible that a developing country such as Mexico has a sluggish rate of labor market adjustment. Robertson (2000) finds that following a wage shock, wages in Mexico, especially in interior states, converge slowly to the new equilibrium.

Table 4
Effect of Chinese import competition on working age population in municipalities, 2000–2010.

Dependent variable: decadal headcount log change								
	All (I)	Males (II)	Females (III)	College (IV)	No college (V)	Age 16–34 (VI)	Age 35–49 (VII)	Age 50–64 (VIII)
	A. Full sample							
ΔExp	−0.626	−0.635	−0.605	−0.381	−0.366	−0.657	−0.927	−0.089
	(0.334)$^{+}$	(0.401)	(0.277)*	(0.55)	(0.285)	(0.463)	(0.281)**	(0.307)
R^2	0.39	0.35	0.42	0.31	0.35	0.29	0.52	0.54
N	1387	1379	1364	1384	1402	1382	1350	1067
	B. Municipalities from border states							
ΔExp	1.011	1.622	0.025	2.168	0.633	0.39	1.458	1.703
	(0.512)*	(0.672)*	(0.523)	(0.852)*	(0.541)	(0.876)	(0.451)**	(1.451)
R^2	0.27	0.27	0.24	0.19	0.24	0.11	0.44	0.52
N	238	238	238	238	238	238	238	214
	C. Municipalities from interior states							
ΔExp	−0.952	−1.009	−0.78	−1.336	−0.434	−1.123	−0.858	−0.533
	(0.308)**	(0.321)**	(0.269)**	(0.435)**	(0.223)$^{+}$	(0.436)*	(0.304)**	(0.217)*
R^2	0.52	0.49	0.52	0.44	0.42	0.46	0.56	0.59
N	1149	1141	1126	1146	1164	1144	1112	853

All regressions include a constant. SE are clustered on state. Models are weighted by start of the period MA share of national working age population with college.

$^{+}$ $p<0.1$.
* $p<0.05$.
** $p<0.01$.

Thus, it would not be surprising to see some within-local labor market adjustments due to the shock to manufacturing industries. In this section, I present the estimated effects of the change in exposure to Chinese imports on employment in the non-manufacturing sector, unemployment, labor-force participation, and labor mobility.

The first variable I consider is workers' mobility. I present in Table 4 the estimated effect of the change in Chinese import competition on working age population. The first column in Panel A shows the estimate using the full sample of municipalities. The coefficient −0.626 is significant at the 6% level. This estimate implies that a standard deviation increase in exposure (0.033) would decrease the municipality's working age population by approximately 2.07 percentage points. Columns II–VIII show estimates for certain demographic groups of interest. Worth of mention are the significant coefficients for female workers and for workers in the 35–49 age group and the insignificant coefficient for male workers and for workers with ages ranging between 16 and 34. First, we would expect the import shock to have a stronger effect on males than on females. Also, it seems reasonable to expect younger workers to migrate more. Thus, the results in Panel A seem slightly puzzling. In an attempt to clear a potentially confounding factor, I divide the full sample into municipalities located in states bordering the U.S.[6] (Table 4, Panel B) and municipalities located in interior states (Table 4, Panel C). Since Mexico has a traditionally large flow of international migrants going to the U.S., it seems plausible for municipalities near the border to experience both a large influx of workers and an increase in Chinese import exposure. The latter due to the intense manufacturing activity in that region. This would create an attenuation bias in the workers' mobility estimate. The results seem to lend some support to this hypothesis. When using only interior municipalities the coefficient in the first column becomes significant at a level of less than 1%, and its absolute size increases by more than 50%. Also, the coefficients for both male workers and 16–34 workers become significant and more sizable than for female workers and 35–49 workers. Panel C also shows significant negative effects for both college

[6] The Mexican states bordering the U.S. are Baja California, Chihuahua, Tamaulipas, Coahuila, Sonora y Nuevo Len. Baja California Sur, although technically not on the border with the U.S., is also included in the sample because of the tax exemptions entitled to it due to its relative proximity with the U.S.

Table 5
Effects of Chinese import competition on employment status in municipalities, 2000–2010.

Dep. Vars: decadal change in headcounts and population share by employment status				
	Mfg Emp (1)	Non-Mfg Emp (2)	Unemployed (3)	Inactive (NILF) (4)
	A. Log change in headcounts			
ΔExp	−1.409	−0.679	−4.6	−0.716
	(0.654)*	(0.5)	(1.044)**	(0.203)**
	B. Change in population shares			
	All			
ΔExp	−0.137	−0.003	−0.041	−0.023
	(0.045)**	(0.043)	(0.024)+	(0.086)
	Males			
ΔExp	−0.288	0.107	−0.08	0.037
	(0.065)**	(0.054)*	(0.036)*	(0.068)
	Females			
ΔExp	0.169	−0.213	−0.002	−0.08
	(0.068)*	(0.088)*	(0.015)	(0.015)
	College			
ΔExp	−0.158	0.151	0.02	0.156
	(0.066)*	(0.1)	(0.025)	(0.099)
	Non-college			
ΔExp	−0.047	−0.042	−0.067	−0.032
	(0.049)	(0.044)	(0.022)**	(0.083)

All regressions include a constant. SE are clustered on state. Models are weighted by start of the period MA share of national working age population with college.

+ $p < 0.1$.
* $p < 0.05$.
** $p < 0.01$.

and non-college workers. Thus, it appears that Chinese import competition had a significant role in the mobility that the Mexican working age population displayed during the decade.

Significant working age population mobility across municipalities must result in corresponding changes within municipalities. We have seen that manufacturing employment decreased, which could explain some of the reduction in the amount of workers in the local labor market. However, it is interesting to see what happened to other worker' status. In particular, I present results for non-manufacturing employment, unemployment, and people who are not in the labor force (NILF). Panel A from Table 5 presents the log change in headcounts for the full sample. This variable tells us what happened to the overall number of workers for employment categories from one period to the next. We can see that the workers' departure from the local labor market was composed not only by the reduction in manufacturing employment, but also by a decrease both in unemployed and in NILF people. There was not a significant change in non-manufacturing employment. Panel B presents changes in population shares due to Chinese import competition for different demographic groups. This variable gives us a better picture of what happened to the employment structure within the local labor market. For the sample including all demographic groups I find that there was a strongly significant decrease in manufacturing employment as a share of working age population. Also, the share of unemployed slightly decreased with a 10% level of significance. In untabulated results, I find that the share of agricultural employment increased accordingly, which explains the lack of significant changes both in the share of non-manufacturing employment and NILF people. For male workers, the decrease in the share of manufacturing employment was more sizable than for the full population. We can also see an increase in the representation of non-manufacturing employment in working age population. For female workers we have the opposite effect. There was an increase in the proportion of manufacturing employment, while the share of non-manufacturing employment decreased. Workers with college education experienced a significant decrease in the share of manufacturing employment with other employment status left unchanged. Finally, workers with no college education saw their share of unemployed significantly, although slightly, decreased. They were not significantly affected otherwise.

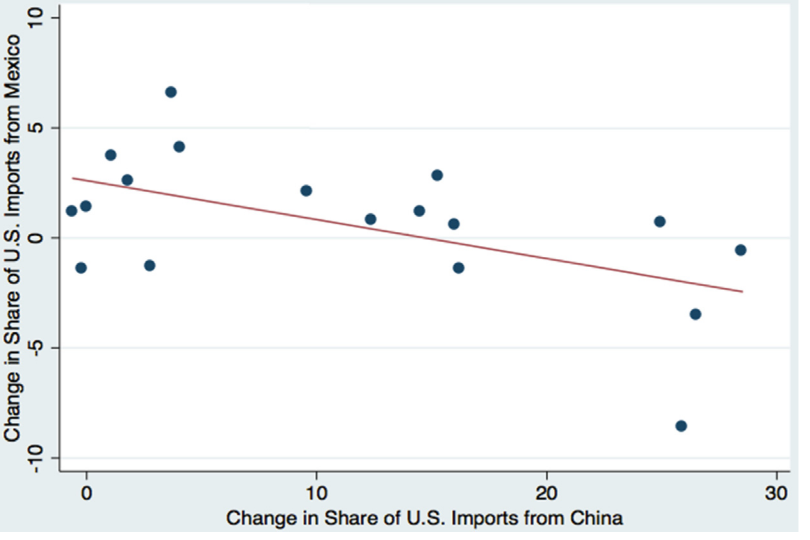

Fig. 4. U.S. imports from Mexico and China.

5.3. *Chinese import competition and trade diversion*

So far I have only studied the direct effect that the Chinese export growth during the 2000s had on the Mexican economy. That is, I have only considered the exogenous increase in domestic competition arising from the larger availability of Chinese goods in Mexican markets. However, this might not be the only significant source of competition. China's share of U.S. imports went from 3.1% to 18.4% during the decade, and as early as 2006 it displaced Mexico as the second largest exporter to the U.S. market. Mexico's share of U.S. imports was 11% between 2000 and 2001, and has not changed much since then, fluctuating between 9.5% and 10.5% until reaching 11% again in 2010. Fig. 4 plots the change in the share of U.S. imports from Mexico and China for three-digit NAICS manufacturing industries between 2000 and 2010. This simple tool allows us to see that throughout the decade there was a significant negative relationship between Mexico and China's growth in manufacturing industries as a share of U.S. imports.

Thus, it seems reasonable to think that perhaps China's competition for U.S. imports has had a negative impact on Mexican manufacturing employment, especially considering that the vast majority of Mexican exports are manufactures, ranging between 93% and 97% of total non-oil exports during the last two decades (Rodriguez-Lopez and Campos-Vazquez, 2011). In order to test the effect Chinese import competition had on Mexican local labor markets by diverting trade between the U.S. and Mexico, I use the following equation:

$$\hat{L}_{Ti} = -\propto \sum_{j} \frac{L_{ij}}{L_{Ti}} \left(\frac{X_{ijm}}{X_{ij}} \frac{M_{mjC}}{E_{mj}} + \frac{X_{iju}}{X_{ij}} \frac{M_{ujC}}{E_{uj}} \right) \hat{A}_{Cj} \tag{7}$$

which captures both the direct effect of China's exports into Mexico and the indirect effect of China-caused trade diversion. It is important to highlight that this form explicitly assumes that no other country is exporting into Mexico, a very reasonable assumption taking into account that by 2010 the U.S. and China represented nearly 98% of Mexican imports (source: UN Comtrade Database).

Table 6
Indirect effect of Chinese import competition on employment in municipalities, 2000–2010.

	Mfg Emp (1)	Non-Mfg Emp (2)
	A. Full sample	
ΔExp (Dom + Intl)	−10.891	−0.007
	(4.130)**	(0.04)
R^2	0.53	0.16
N	1387	1387
	B. Border states	
ΔExp (Dom + Intl)	−18.331	0.05
	(5.041)**	(0.039)
R^2	0.61	0.3
N	209	209

All regressions include a constant. SE are clustered on state. Models are weighted by start of the period MA share of national working age population with college. Border States exclude 22 municipalities with preferential Value Added Tax treatment.
\+ $p < 0.1$.
* $p < 0.05$.
** $p < 0.01$.

In order to empirically estimate Eq. (7) I use (i) sales in the U.S. from region i in industry j (X_{iju});[7] (ii) U.S. imports from China in industry j (M_{ujC});[8] and (iii) total sales in the U.S. in industry j (E_{uj}), proxied by total shipments in the U.S. in industry j. Table 6 summarizes the results for manufacturing and non-manufacturing employment. In Panel A, I present the estimate of the average effect that exposure to both domestic and international Chinese competition had on employment for the full sample of municipalities. The coefficient −10.89 in the proportion of manufacturing employment is significant at a level of less than 1% and implies that an increase of one standard deviation in exposure decreased the proportion of manufacturing employment by 0.6 percentage points. Although the absolute value of the coefficient decreased with respect to the one with only the domestic exposure index, the greater variation in the full exposure index (domestic and international) results in a stronger effect by standard deviation. Fig. 5 sketches the specification for the change in manufacturing employment in Panel A. There was no significant effect of indirect exposure on non-manufacturing employment.

In Panel B, I present results for border states only. Since border states are home of nearly 80% of maquiladora establishments (U.S. General Accounting Office, 2003), it is interesting to assess whether local labor markets in this region were more affected by Chinese import competition in the U.S. market. Maquiladoras export almost 100% of their production to the U.S., and it is plausible that municipalities that rely more heavily on this type of industry are more affected by competition than municipalities from interior states. The coefficient −18.331 is much larger in absolute value than the coefficient with the full sample. This estimate implies that an increase of a standard deviation in exposure would decrease manufacturing employment by 1.03 percentage points, an effect more than 60% stronger than the estimated for the full sample. This shows that municipalities with a high share of total maquiladora establishments have a higher dependence on their sales abroad, and thus, are more sensitive to Chinese import competition in the U.S. market. Fig. 6 depicts these results.

5.4. Wages

Table 7 shows the effects of Chinese import competition on wages. The dependent variable for the analysis is the log of weekly wage. Column 1 shows the estimated effect for the full sample, suggesting a small positive effect of competition on wages. The coefficient 0.74 implies that an increase of one

[7] I am restricted by the lack of export data by municipality. To proxy for X_{iju} I use maquiladora sales per municipality, per industry. Maquiladoras export to the U.S. nearly 100% of their output. I am implicitly assuming that changes in exports from non-maquiladora firms do not have a significant effect on employment.

[8] These imports are instrumented by imports to a set of developed countries: Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland.

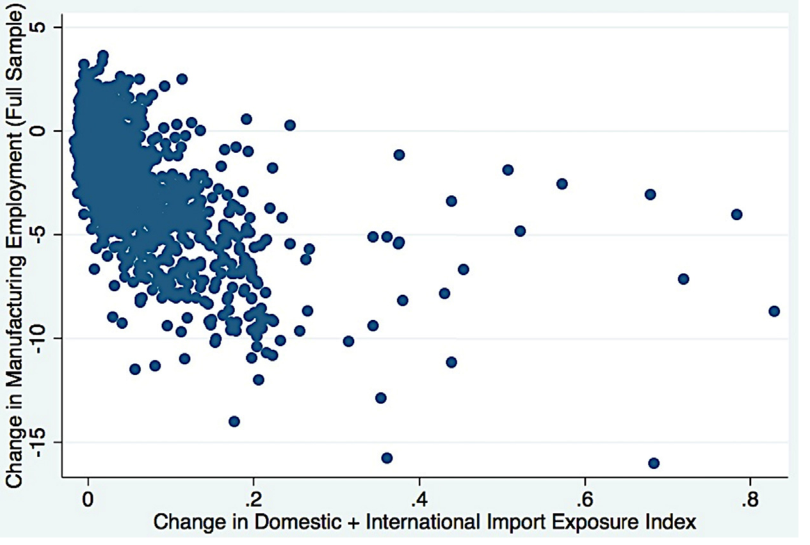

Fig. 5. Indirect effect on manufacturing employment 2000–2010.

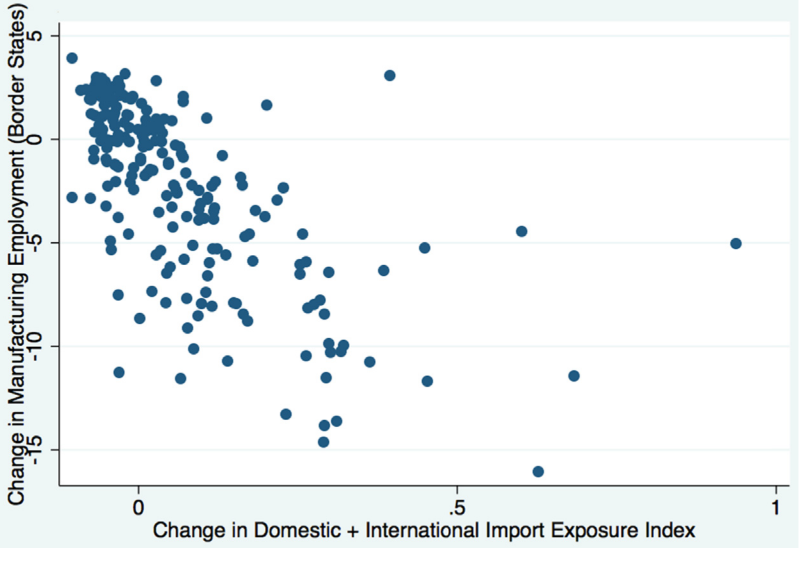

Fig. 6. Indirect effect on border manufacturing employment 2000–2010.

standard deviation in exposure to competition (0.033) would increase weekly wages by approximately 2.4 percentage points. This effect is stronger for male workers (coefficient 0.92) than for female workers (0.44). Also, the increase in Chinese competition does not seem to have a significant effect by industry or level of education. The non-negative effect of Chinese import competition on wages combined with a strong decline in manufacturing employment is not an uncommon finding. Autor et al. (2013) and Edwards and Lawrence (2010), both for the U.S. case, find that, although import exposure reduces manufacturing employment, no negative wage effects on U.S. workers in import competing manufacturing industries are observed. Several theories have been put forward to explain this pattern. One possibility is that firms react to competition by innovating in their productive processes, allowing them

Table 7
The effect of Chinese import competition on wages.

Dep. Var: decadal change in avg. log weekly wage							
	All (1)	Manuf (2)	Non-manuf (3)	Males (4)	Females (5)	College (6)	No college (7)
ΔExp	0.74	0.769	0.529	0.917	0.442	0.475	0.367
	(0.308)*	(0.572)	(0.342)	(0.431)*	(0.226)*	(0.409)	(0.34)
R^2	0.62	0.47	0.6	0.56	0.62	0.59	0.44
N	1387	848	1387	1379	1364	1384	1402

All regressions include a constant. SE are clustered on state. Models are weighted by start of the period MA share of national working age population with college. Specification includes beginning-of-the-period log weekly wage as control.
+ $p < 0.1$.
** $p < 0.01$.
* $p < 0.05$.

to increase their productivity and pay higher wages (Bloom, Draca, & Van Reenen, 2012). Of course, not all firms are capable of undergoing such transition, and jobs end up being lost, explaining the reduction in manufacturing employment. On a similar note, downward rigid wages can be explained by only the most productive workers retaining their jobs, biasing thus the estimate against finding a wage effect. Alternatively, wage effects could be masked by shifts in employment compositions (Autor et al., 2013), which seems consistent with the high degree of mobility that workers displayed during the period (and the subsequent reduction of regional labor supply) and the increase in agricultural employment. It is quite surprising to find that, within local labor markets, general equilibrium effects operate efficiently by dissipating the negative effect on manufacturing employment through a combination of out-of-the-region mobility, across-sector mobility, and firm productivity-enhancing behavior. This contrasts with the persistence of the negative shock on manufacturing employment, which is not dissipated across local labor markets.

5.5. *Falsification test*

To ensure that the obtained results are product of period-specific changes in exposure to Chinese import competition and not just long-running changes in regional employment, I regress the change in manufacturing employment between 1990 and 2000 on the change in regional trade exposure (both direct and indirect) between 2000 and 2010. If it is the case that Mexican local labor markets are being affected by exposure to Chinese trade, then changes in manufacturing employment during the 1990s, a period when Chinese imports were relatively low, should bear a weak relationship with future changes in trade exposure. A statistically insignificant positive coefficient (13.95, with a p-value of 0.23) lends its support to the hypothesis that the relationship between declining manufacturing employment and increasing Chinese trade exposure was not present in the period before the Chinese trade shock.

5.6. *Policy implications*

In theory, Mexico gains from trading internationally. As we have seen, trade with China had a few negative effects on regions more exposed to import competition, such as reduced employment in manufactures and an increased level of workers' mobility, which has social and economic costs of its own. However, income growth in regions with expanding exports, reduced cost of intermediate inputs, and an increased product variety for consumers should ensure that the overall balance of the trade relationship is positive. What we obtain from the analysis is that, in spite of these gains, the increase of trade volume between Mexico and China had distributional consequences that affected unevenly those regions more vulnerable to increases in Mexico's import penetration from China. What lesson can we learn from this? First, if the increase in imports of labor-intensive goods is affecting the domestic market both directly through domestic competition and competition abroad, it is imperative to continue the process of augmenting the value added and differentiation of the goods produced. Government support on these developments must be a component of the social net aimed at protecting trade-exposed regions. A complementary approach could be to mimic the U.S. and implement

Trade Adjustment Assistance (TAA) benefits, which are targeted specifically at individuals who lost employment due to trade competition. The first approach is a more productive and preventive measure, although it can be considered a more medium-to-long term solution, while the second one helps to correct and diminish the uneven negative consequences of trade almost immediately. A combination of the two could ensure that the negative distributional consequences of trade are kept to a minimum as globalization through trade continues to expand and to become more important in determining labor market outcomes in all economies.

6. Conclusions

China has become a protagonist in the economic world stage. Its dramatic growth during the last two decades has had an impact in most countries in the world. Estimating the full impact of this development is perhaps now more than ever an important exercise for all affected economies. In this particular case, I estimate the effect of Chinese trade growth on the Mexican local labor markets. Characteristics such as its dependence on labor-intensive industries, its closeness to the United States, and its classification as a developing country make of Mexico a very interesting case of study. I find that the increase in exposure to Chinese competition due to the growth in China's exports decreased the employment share in manufactures for the average Mexican local labor market. My baseline estimates suggest that an increase of one standard deviation in exposure decreased the manufacturing share of employment by 0.45 percentage points; this effect was found to be larger for local labor markets with high exposure to Chinese competition in the U.S. market, showing that there was a significant negative indirect effect from China's trade growth. Additionally, I find that there was a significant degree of workers' mobility caused by the negative trade shock. A standard deviation increase in exposure is correlated with a two-percentage point reduction in a municipality's working age population. This effect is even larger and more significant for municipalities located in states that are not on the border with the U.S. Changes in employment status' composition were mostly characterized by a reduction in the manufacturing employment share, an increase in the agricultural employment share, and mixed results by gender and education in unemployment and NILF-people shares. No negative significant effects were found for the non-manufacturing employment share, except for female workers. Wages were not negatively affected by the increase in exposure, and under some specifications I find that there was a positive effect.

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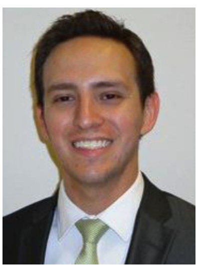

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